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CST1510

Memo No. LABS

Date 30 / 09 / 2026

Conditional Statements

→ Always press tab after an 'if' statement

otherwise it won't print

```
value = 23.7
```

```
limit = 20
```

```
if value > limit:
```

```
    print ("OVER")
```

this is incorrect

```
value = 23.7
```

```
limit = 20
```

```
if value > limit:
```

```
    print ("OVER")
```

this is correct.



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W02 - Lab

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D1 → done in class

D2

limit = 50 → stores value

print (limit == 50) → asks a question

print (limit == 60) → asks a question

D3

attempts = 4

max_attempts = 3

is_locked = attempts >= max_attempts

print (is_locked)

print (type(is_locked))

D4

percent = 95

print (percent >= 90 and percent < 100)

print (percent < 50 or percent > 90)

print (not (percent > 90))



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W02 - Lab

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D4

percent = 40

print (percent >= 90 and percent < 100)

print (percent < 50 or percent > 90)

print (not (percent > 90))

predicted results : False, True, True

D5

value = 120

limit = 100

if value > limit :

print ("OVER LIMIT")

value = 80

limit = 100

if value > limit :

print ("OVER LIMIT")

→ ~~80 is not greater~~ 80 is not greater
than 100.



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W O J - Lab

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D6

value = 87 $\longrightarrow$ if 120 was the value,

limit = 100 it would've printed

if value > limit : OVER LIMIT.

print ("OVER LIMIT")

else :

print ("OK")

D7

value = 87

limit = 100

percent = (value / limit) * 100 > for value 87 this

if percent >= 100 would output OK

print ("OVER LIMIT") > for value 95 this

elif percent >= 90 : would output WARNING

print ("WARNING") > for value 120 this

else would output OVER

print ("OK") LIMIT



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D8

1. # the code would print ~~check~~ Check finished.

4. # nothing prints because value > limit is false

and (both print statements are inside the if.)

D9

count = 1

while count <= 10 :

print(count)

count += 1

D10

broken code:

correct code:

count = 1

count = 1

while count <= 5:

while count <= 5:

print(count)

print(count)

inside the loop it

count += 1

only prints count 5

count starts at 1, it prints



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W102 - Lab

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Q11

total = 0

while True:

number = input("Enter a number (or done to finish):")

if number == "done":
break

total += float(number)
print("Total: ", total)

Q12

x = 1

while x <= 10:

if x == 5:

x = x + 1

continue

print(x)

x = x + 1



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W02 Lab

D13

```
for i in range(5):  
    print(i)
```

} 0, 1, 2, 3, 4

```
for i in range(1, 6):  
    print(i)
```

} 0, 1, 2, 3, 4, 5

D14

even numbers

```
for i in range(2, 21, 2):  
    print(i)
```

21 is used because the stop number isn't included

this 2 at the end means it starts at 2 and increases by 2 each time

```
for i in range(10, 0, -1):  
    print(i)
```

start

go down by 1 each time

stop before 0

Note:

```
for i in range(3):  
    print(i)
```

range(start, stop, step) # 0, 1, 2

stop is never included

```
for i in range(2, 5):  
    print(i)
```

2, 3, 4

```
for i in range(10, 0, -3):  
    print(i)
```

10, 7, 4, 1



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W02_Lab

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D15

```
code = "A7x"
```

```
for position, fruit in enumerate(["apple", "pear"], start=1):  
    print("Fruit", position, ":", fruit)
```

D15

```
code = "A7x"
```

```
for position, character in enumerate(code, start=1):
```

```
    print(f"Character {position} : {character}")
```

D16

```
for i in range(3):
```

```
    number = input("Enter a number: ")
```

```
    print(number)
```

} for
we know
the number
of times it needs
to run.

```
password = ""
```

```
while password != "letmein":
```

```
    password = input("Enter password: ")
```

```
    print("Welcome")
```

} while
we don't know
the number of
times the
user needs.



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W02 - Lab

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C1

```
for i in range(3):  
    value = int(input("Enter a value: "))  
    limit = int(input("Enter a limit: "))  
  
    if value > limit:  
        print("OVER LIMIT")  
    else:  
        print("OK")
```

C2

```
correct_password = "python123"  
  
tries = 0  
  
while tries < 3:  
    password = input("Enter password: ")  
    tries = tries + 1  
  
    if password == correct_password:  
        print("Access granted")  
        break  
    elif tries == 3:  
        print("Account Locked")
```



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WEEK 2 NOTES

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Comparisons

Comparison operators check whether two ~~ba~~ values are equal, different, bigger, or smaller. They give a result of True or False.

$== \rightarrow$ equal to $> \rightarrow$ greater than

$!= \rightarrow$ not equal to $<= \rightarrow$ less than or equal to

$< \rightarrow$ less than $>= \rightarrow$ greater than or equal to

$= \rightarrow$ assigns a value, while $==$ checks if two values are equal.

Combining Conditions

You can combine multiple conditions using and, or and not.

and $\rightarrow$ both conditions are true (both)

or $\rightarrow$ at least one condition is true (either)

not $\rightarrow$ reverse True / False (opposite)



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WEEK 2 NOTES

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Decisions

These let Python make decisions based on conditions.

if → checks the first condition.

elif → checks another condition if the previous one was false.

else → runs if none of the conditions were true.

Python runs the first branch that is

true, then skips the others.

Indentation

Indentation means leaving spaces at the beginning of a line. It tells Python which lines belong inside an if statement or loop.

```
age = 16
```

```
if age >= 16
```

```
    print("You can apply")
```




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while Loops

A while loop repeats code while a condition eventually

is True. The condition needs to become false,

otherwise the loop can continue forever.

count = 1

while count <= 3:

 print(count)

 count = count + 1

} output: 1
 2
 3

while True creates a loop that keeps running

until ~~something~~ ^{something} stops it.

break → stops the loop

continue → skips the current iteration and

goes to the next one.



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for loops

A for loop is useful when you know how many times you want to repeat something or want to go through items one by one.

range() creates a sequence of numbers.

- range(stop) → starts at 0
- range(start, stop, step) → choose where to start and how much to increase by
- The stop number is NOT included.

for i in range(3): = for i in range(2, 8, 2):

print(i) = print(i)
0, 1, 2 2, 4, 6.

enumerate() lets you get both the position/
index and the item when looping through something



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for or while

Use for when you know how many times you want something to happen.

for i in range(5):

print("Hi")

Use while when you don't know exactly how many times it needs to repeat and instead want it to continue until a condition changes.

password = ""

while password != "hello":

password = input("Enter password:")